

EGSE Documentation

The Electronic Ground Support Equipment consists of a user's host computer, to run and host YAMCS and to run programs to handle and transport incoming data to YAMCS.

The current version of the software contains a configured build of YAMCS, a custom created XTCE database made with the Opus2 Tool and a spacecraft simulator program to represent simple behaviour of a spacecraft.

All of this software is available on Github at the link: <https://github.com/KISPE-Space/mbep-cdr-quickstart>

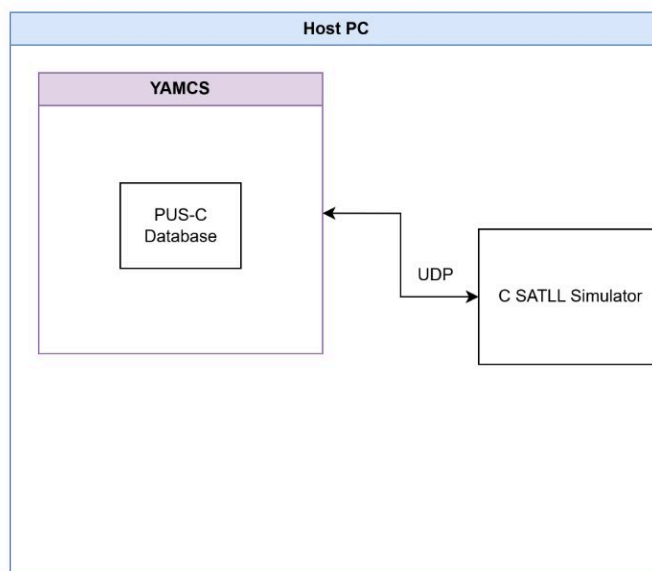
This repository contains the aforementioned software and this documentation.

The guide includes steps for the user to:

- Download and run the YAMCS instance tailored for the MBEP CDR
- Run the spacecraft simulator program
- View incoming telemetry and send commands to the simulator

Setup

this software configuration assumes the physical setup as shown below:



All that is required is a host PC as this example is primarily software based. This software and setup has been tested and ran on Ubuntu systems and is known to work. YAMCS can be run on windows but it has not been tested in this configuration.

YAMCS

Install Script

Provided in the repo in `/scripts` are three bash scripts. The script `install-yamcs.sh` installs the adapted version of YAMCS to work for the CDR.

To run the install script, in the same directory as the `.sh` file, in a terminal enter:

```
bash install-yamcs.sh
```

or it can be run with

```
chmod +x install-yamcs.sh # Only ever need to run once  
./install-yamcs.sh
```

This will default to installing the GitHub repo to the user's `/home/USERNAME/egse` directory. This is default but can be adapted by editing the script itself. The script will then extract the required files and YAMCS is ready to run.

Running YAMCS

YAMCS can be run with the script `run-yamcs.sh` assuming the previous install script has been run. Note if you adapt the install script to install to a different location, the run script will also need to be updated.

```
bash run-yamcs.sh
```

To run YAMCS alongside the `satll-simulator` run the command:

```
bash egse_run_simulator.sh
```

Where the YAMCS log output will be displayed in a new terminal alongside the simulator output.

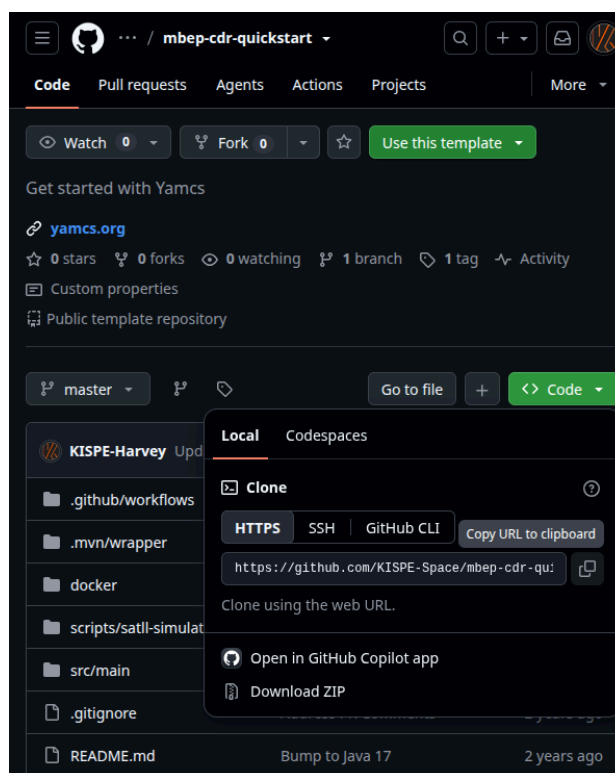
Manually Installing YAMCS

YAMCS has been precompiled and available at the repo mentioned. The files provided also allow a user to customise and recompile the project if desired.

The YAMCS instance is packaged as `mbep-cdr-yamcs-0.0.1-CDR-bundle.tar.gz` which can be extracted and ran.

The code can be downloaded by selecting the `Code` button and downloading as a `.zip` which can be decompressed or by cloning the git repo using the HTTPS link:

`https://github.com/KISPE-Space/mbep-cdr-quickstart.git` .



Once downloaded the `.tar.gz` can be extracted. Extract the `.tar.gz` file to your machine.

```
tar -xzf mbep-cdr-yamcs-0.0.1-CDR-bundle.tar.gz -C /PATH/ON/COMPUTER
```

Then enter the directory of the extracted file, where running the `ls` command should show a directory called `bin`. If `bin` is visible run the command:

```
./bin/yamcsd
```

on Windows:

```
bin\yamcsd.bat
```

This command should then show the following outputs:

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS MEMORY XRTOS
harvey@harvey-T480s:~/github/kispe/mbep-cdr-quickstart/target$ tar -xzf mbep-cdr-yamcs-0.0.1-CDR-bundle.tar.gz -C .
harvey@harvey-T480s:~/github/kispe/mbep-cdr-quickstart/target$ cd mbep-cdr-yamcs-0.0.1-CDR/
harvey@harvey-T480s:~/github/kispe/mbep-cdr-quickstart/target/mbep-cdr-yamcs-0.0.1-CDR$ ./bin/yamcsd
23:05:02.233 global [I] RdbStorageEngine Creating or loading tablespace_global
23:05:02.237 global [I] Tablespace [global] Creating database at yamcs-data/global.rdb
23:05:02.272 global [I] Tablespace [global] Loaded 0 tables for instance_global
23:05:02.274 global [I] YamcsServer Loading service HttpServer
23:05:02.827 global [I] RdbProtobufDatabase Created new protobuf db tbsIndex: 5
23:05:03.122 global [I] Directory Saving new user [name=admin, id=6]
23:05:03.285 global [I] YamcsServer Loading online instance 'mbep-cdr-yamcs'
23:05:03.350 global [I] XtceStaxReader Parsing XTCE file mdb/mbep-cdr-satll.xml
23:05:03.486 global [I] XtceStaxReader XTCE file parsing finished, loaded: 127 parameters, 50 tm containers, 92 commands
23:05:03.662 global [I] MdbFactory Serialized database saved locally
23:05:03.662 global [I] RdbStorageEngine Creating or loading tablespace mbep-cdr-yamcs

23:05:04.134 global [I] YamcsServer Yamcs 5.13.0, build 8bf5af6fe227bbf8ead15e60644b7ddb345d623
23:05:04.468 mbep-cdr-yamcs [I] YamcsServerInstance Awaiting start of service XtceTmRecorder
23:05:04.469 mbep-cdr-yamcs [I] YamcsServerInstance Awaiting start of service ParameterRecorder
23:05:04.469 mbep-cdr-yamcs [I] YamcsServerInstance Awaiting start of service AlarmRecorder
23:05:04.469 mbep-cdr-yamcs [I] YamcsServerInstance Awaiting start of service EventRecorder
23:05:04.469 mbep-cdr-yamcs [I] YamcsServerInstance Awaiting start of service ReplayServer
23:05:04.469 mbep-cdr-yamcs [I] YamcsServerInstance Awaiting start of service SystemParametersService
23:05:04.469 mbep-cdr-yamcs [I] YamcsServerInstance Awaiting start of service ProcessorCreatorService
23:05:04.469 mbep-cdr-yamcs [I] YamcsServerInstance Awaiting start of service CommandHistoryRecorder
23:05:04.469 mbep-cdr-yamcs [I] YamcsServerInstance Awaiting start of service ParameterArchive
23:05:04.469 mbep-cdr-yamcs [I] YamcsServerInstance Awaiting start of service ParameterListService
23:05:04.469 mbep-cdr-yamcs [I] YamcsServerInstance Awaiting start of service TimelineService
23:05:04.469 mbep-cdr-yamcs [I] YamcsServerInstance Awaiting start of service ParameterRetrievalService
23:05:04.473 global [I] YamcsServer Yamcs started in 2953ms. Started 1 of 1 instances and 13 services
23:05:04.475 global [I] WebPlugin Website deployed at http://localhost:8090

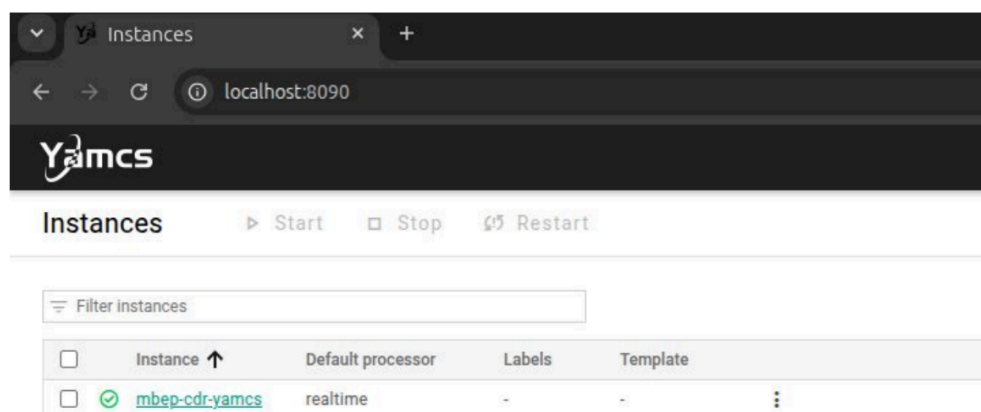
```

This states, due to configuration of this instance, YAMCS is hosted on the user's computer at <http://localhost:8090>.

Connecting to YAMCS

As mentioned YAMCS runs as a server on the machine, running the command `./bin/yamcsd` runs this service. As shown in Image 2, to access the server/web interface visit <http://localhost:8090> or can be accessed by `ctrl + click` the terminal output.

This should bring up the YAMCS web interface. This is the default screen:



Viewing Parameters

YAMCS parses the database which contains all of the expected datatypes and parameters. They can be seen in the Telemetry/Parameters section.

Yamcs

mbep-cdr-yamcs

Home

Links

Telemetry

Packets

Parameters

Parameter lists

Displays

Algorithms

Events

Alarms

Commanding

Procedures

Timeline

Activities

Mission database

Archive browser

Parameters

ASW

Search by name

Any type

Any source

Columns

Name	Type	Data source	Value	Description
17_4_application_process_id	enumeration	Telemetered	test_apid	-
1_10_application_process_id	enumeration	Telemetered	test_apid	-
1_10_failure_notice_code	enumeration	Telemetered	unknown_routing_failure	-
1_10_packet_sequence_count	integer	Telemetered	-	-
1_10_packet_type	enumeration	Telemetered	Telemetry	-
1_10_packet_version_number	enumeration	Telemetered	CCSDS_packet	-
1_10_secondary_header_flag	binary	Telemetered	-	-
1_10_sequence_flags	enumeration	Telemetered	standalone_packet	-
1_1_application_process_id	enumeration	Telemetered	test_apid	-
1_1_packet_sequence_count	integer	Telemetered	-	-
1_1_packet_type	enumeration	Telemetered	Telemetry	-
1_1_packet_version_number	enumeration	Telemetered	CCSDS_packet	-
1_1_secondary_header_flag	binary	Telemetered	-	-
1_1_sequence_flags	enumeration	Telemetered	standalone_packet	-
1_7_application_process_id	enumeration	Telemetered	test_apid	-

For this example the added parameters have the prefix **onboard** :

Yamcs

mbep-cdr-yamcs

Home

Links

Telemetry

Packets

Parameters

Parameter lists

Displays

Algorithms

Events

Alarms

Commanding

Procedures

Timeline

Activities

Mission database

Archive browser

Parameters

ASW

onboard

Any type

Any source

Columns

Name	Type	Data source	Value	Description
onboard_cmd_acc_cnt	integer	Telemetered	-	-
onboard_cmd_rcv_cnt	integer	Telemetered	-	-
onboard_day	integer	Telemetered	-	-
onboard_fw_db_ver	integer	Telemetered	-	-
onboard_hour	integer	Telemetered	-	-
onboard_minute	integer	Telemetered	-	-
onboard_sec_id	integer	Telemetered	-	-
onboard_second	integer	Telemetered	-	-

Viewing Available Commands

Commands can be sent from YAMCS and are avaihle in the **Commanding/Send a command** section. Commands can be configured to invoke specific actions and can request acknowledgements from the simulator.

Yamcs

mbep-cdr-yamcs

Home

Links

Telemetry

Algorithms

Events

Alarms

Commanding

Send a command

Command history

Queues

Procedures

Timeline

Activities

Mission database

Archive browser

Send a command

ASW

Search by name

Columns

Name	Significance
acquire_data_from_logical_devices_base	-
acquire_data_from_logical_devices_n1	-
acquire_data_from_logical_devices_n2	-
acquire_data_from_logical_devices_n3	-
acquire_data_from_logical_devices_n4	-
acquire_data_from_physical_devices_base	-
acquire_data_from_physical_devices_n1	-
acquire_data_from_physical_devices_n2	-
acquire_data_from_physical_devices_n3	-
acquire_data_from_physical_devices_n4	-
respond_parameters_to_a_housekeeping_parameter_report_structure	-
create_a_housekeeping_parameter_report_structure	-
delete_housekeeping_parameter_report_structures_base	-
delete_housekeeping_parameter_report_structures_n1	-

Configuring a command:

Send a command

1. Select command

2. Configure command

3. View report

Command

System

Significance

distribute_on_off_device_commands_base

/ASW

-

Arguments

source_id

unsigned integer

2_1_N

unsigned integer

tc_version

unsigned integer

2

ack_flags

unsigned integer

0

Advanced options

Comment

Cancel

Send

for the ack flags:

	Acceptance Request	Start Request	Progress Request	Completion Request
Successful	TM[1,1]	TM[1,3]	TM[1,5]	TM[1,7]
Unsuccessful	TM[1,2]	TM[1,4]	TM[1,6]	TM[1,8]
As per the SATLL ICD, the simulator only supports TM[1,1] and TM[1,7] responses (and corresponding unsuccessful).				

For both an **Acceptance Request** and a **Completion Request** . The **ack_flags** should be :

Acceptance Request	Completion Request	Binary	Decimal
0	0	0000	0
1	0	1000	8
0	1	0001	1
1	1	1001	9

A history of Command send is saved and can be inspected by the user.

Running the simulator

The Github repo also contains a precompiled version of the simulator for ease of running, along with the source code and make files needed to recompile the code.

To run the simulator navigate to the `/scripts/satll-simulator` directory. To run the simulator run the command `./satll-sim`. This should be run in a terminal which will output to the terminal the data being sent.

```
harvey@harvey-T480s:~/github/kispe/mbe-cdr-quickstart/target/mbe-cdr-yamcs-0.0.1-CDR$ ./bin/yamcsd
09:09:37.716 mbe-cdr-yamcs [1] YamcsServerInstance Awaiting start of service ReplayServer
09:09:37.716 mbe-cdr-yamcs [1] YamcsServerInstance Awaiting start of service SystemParametersService
09:09:37.716 mbe-cdr-yamcs [1] YamcsServerInstance Awaiting start of service ProcessorCreatorService
09:09:37.716 mbe-cdr-yamcs [1] YamcsServerInstance Awaiting start of service CommandHistoryRecorder
09:09:37.716 mbe-cdr-yamcs [1] YamcsServerInstance Awaiting start of service ParameterArchive
09:09:37.717 mbe-cdr-yamcs [1] YamcsServerInstance Awaiting start of service ParameterListService
09:09:37.717 mbe-cdr-yamcs [1] YamcsServerInstance Awaiting start of service TimelineService
09:09:37.717 mbe-cdr-yamcs [1] YamcsServerInstance Awaiting start of service ParameterRetrievalService
09:09:37.719 _global [1] YamcsServer Yamcs started in 2186ms. Started 1 of 1 instances and 13 services
09:09:37.720 _global [1] WebPlugin Website deployed at http://localhost:8090
[1]
```

```
harvey@harvey-T480s:~/github/kispe/mbe-cdr-quickstart/scripts/satll-simulator$ ./satll-sim
Sending: 0801C001001D21031900000000056A3D00010001001A000B0012002400
6900000000755A
Sending: 0801C003001D21031900020000056A3D00010001001A000B0012002600
690000000077CC
Sending: 0801C005001D21031900040000056A3D00010001001A000B0012002800
6900000000E30B
```

Live Telemetry

Live telemetry should now be coming into YAMCS via the UDP port. Live telemetry and packets can be viewed in the `Telemetry/Packets` tab. Packets are time stamped and the raw data can be extracted and viewed:

Telemetry

Packets

Parameters

Parameter lists

Displays

Algorithms

Events

Alarms

Commanding

Procedures

Timeline

Activities

Query

Saved queries

Last hour

Search packets

Showing data from the last hour ending at 2026-08-13 08:13:38.407 UTC (Mission Time)

Packet name	Generation time	Reception time	Link	Size	
/ASW/housekeeping_parameter_report_HPRS_Housekeeping_1	2026-08-13 08:13:36.492 UTC	2026-08-13 08:13:36.491 UTC	udp-in	36 bytes	Extract
/ASW/housekeeping_parameter_report_HPRS_Housekeeping_1	2026-08-13 08:13:34.492 UTC	2026-08-13 08:13:34.491 UTC	udp-in	36 bytes	Extract
/ASW/housekeeping_parameter_report_HPRS_Housekeeping_1	2026-08-13 08:13:32.492 UTC	2026-08-13 08:13:32.491 UTC	udp-in	36 bytes	Extract
/ASW/housekeeping_parameter_report_HPRS_Housekeeping_1	2026-08-13 08:13:30.492 UTC	2026-08-13 08:13:30.491 UTC	udp-in	36 bytes	Extract
/ASW/housekeeping_parameter_report_HPRS_Housekeeping_1	2026-08-13 08:13:28.492 UTC	2026-08-13 08:13:28.491 UTC	udp-in	36 bytes	Extract
/ASW/housekeeping_parameter_report_HPRS_Housekeeping_1	2026-08-13 08:13:26.492 UTC	2026-08-13 08:13:26.491 UTC	udp-in	36 bytes	Extract
/ASW/housekeeping_parameter_report_HPRS_Housekeeping_1	2026-08-13 08:13:24.492 UTC	2026-08-13 08:13:24.491 UTC	udp-in	36 bytes	Extract
/ASW/housekeeping_parameter_report_HPRS_Housekeeping_1	2026-08-13 08:13:22.492 UTC	2026-08-13 08:13:22.491 UTC	udp-in	36 bytes	Extract

← /ASW/housekeeping_parameter_report_HPRS_Housekeeping_1 / 2026-08-13T08:13:36.492Z / 134332649		Expand all	Collapse all	2026-08-13 08:17:01.407 UTC	
		Hex view			
		0000: 0801 c0e9 001d 2103 1900 e800 0005 6a3d]= 0010: 0001 0001 001a 000b 0016 001c 0069 00001.. 0020: 0000 7ca3			
Loc	Bits	Entry	Type	Raw value	
✓ /ASW/CCSDSPacket			container		
0	16	> CCSDS_Packet_ID	aggregate		
16	16	> CCSDS_Packet_Sequence	aggregate		
32	16	CCSDS_Packet_Length	integer	29	
✓ /ASW/CCSDSPUTelemetryPacket			container		
48	88	> PUS_Data_Field_Header	aggregate		
✓ /ASW/housekeeping_parameter_report_base			container		
136	8	3-25_housekeeping_parameter_report_structure_ID	integer	1	
✓ /ASW/housekeeping_parameter_report_HPRS_Housekeeping_1			container		
144	16	onboard_fw_db_ver	integer	1	
160	16	onboard_day	integer	26	
176	16	onboard_hour	integer	11	
192	16	onboard_minute	integer	22	
208	16	onboard_second	integer	28	
224	16	onboard_sc_id	integer	105	
240	16	onboard_cmd_rcv_count	integer	0	
256	16	onboard_cmd_acc_cnt	integer	0	

The same parameter viewed in the parameters tab will show the latest updated value. The status and activity of the input ports to YAMCS can be viewed in the **Links** tab:

Links

EnableDisableMore

Links

Action log

Filter links

Columns

	Name	In	Out	Detail	
☐	 udp-in	264	0	OK, receiving on 10011	⋮
☐	 udp-out	0	0	OK, sending to localhost:10022	⋮

 OK

 OK, activity

 Error

 Disabled

NOTE

Please note this is still an experimental release of the software as a proof of concept using YAMCS with SATLL. Any issues or feedback or useful ideas please contact hnixon@kisper.co.uk . This software and setup has been tested and ran on Ubuntu systems and is known to work. YAMCS can be run on windows but it has not been tested in this configuration.

